

Conversational AI Meets Documents

Revolutionizing PDF Interaction with GenAI

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Abstract—Navigating large PDF documents manually or using basic keyword-based tools can be tedious, especially in fields like academic research, legal analysis, and healthcare, where accurate and timely information retrieval is crucial. To address this, a system integrates Generative AI (GenAI) with a Retrieval-Augmented Generation (RAG) framework, allowing users to interact with PDF content through natural language queries, making the search process faster and more intuitive. For example, querying a legal document for payment terms provides clear answers, helping users navigate complex documents. Traditional keyword-based methods lack contextual understanding, a limitation overcome by the system, which transforms document content into vector embeddings using models like BERT or GPT. These embeddings capture semantic details for precise retrieval from a vector database. Researchers can quickly extract relevant sections of academic papers without manually scanning the entire document. The system processes content into vector embeddings, stores them in a vector database, and generates coherent responses using a large language model when a user submits a query. This approach simplifies tasks like analyzing healthcare reports. Additionally, the system features session management, ensuring continuity across multiple queries. This allows users, such as legal professionals, to maintain context when reviewing a contract, enabling coherent responses to various queries about clauses or penalties. Future enhancements could include supporting data formats like charts, tables, or images, further expanding the system's versatility. By addressing inefficiencies in traditional navigation and introducing advanced features like session continuity, this system revolutionizes document workflows in research, legal analysis, and healthcare. It sets a new standard for conversational AI tools, transforming how users interact with PDF documents.

Keywords—Generative AI, Retrieval-Augmented Generation, Vector Database, Semantic Search, Document Embedding, Knowledge Extraction

I. INTRODUCTION

Generative AI (GenAI) has significantly advanced the field of artificial intelligence by enabling machines to produce human-like responses and generate content from vast amounts of data. One of its transformative applications is in enhancing how users interact with static documents like PDFs, traditionally navigated through manual searches and scrolling [1]. These methods are time-consuming, especially when dealing with large volumes of information in domains such as legal, academic, and healthcare fields. As a solution,

the integration of Retrieval-Augmented Generation (RAG) frameworks with GenAI introduces a new paradigm where static documents become interactive, allowing users to pose natural language queries and receive precise, contextually relevant answers from the content [2].

This article explores the innovative use of GenAI within the RAG framework to facilitate conversational engagement with PDF documents. By converting documents into vector embeddings using models like BERT or GPT, the system enables efficient semantic searches and real-time response generation [3]. The implementation of session management is crucial to maintaining context over multiple interactions, ensuring coherent conversations. This advancement not only saves time but also enhances user experience by simplifying document navigation and allowing for more natural, dynamic exchanges [4]. The introduction of GenAI in document interaction has significant implications for fields that rely heavily on document analysis, and this paper outlines the architecture, challenges, and potential of this technology.

II. LITERATURE REVIEW

Early document navigation relied on keyword-based searches and hierarchical systems, using techniques like Boolean retrieval and TF-IDF [5]. While effective for structured data, these methods struggled with unstructured data and required users to sift through results manually, making it time-consuming to extract insights. Additionally, they lacked the ability to maintain context across sessions, complicating multi-step navigation for lengthy, technical documents [6]. This highlighted the need for intelligent, context-aware systems that enable dynamic, real-time exploration, shifting away from static retrieval methods.

Generative AI has transformed document interaction by introducing advanced models capable of understanding and generating human-like text based on contextual cues. Technologies such as OpenAI's GPT, Google's Gemini, and HuggingFace's BERT have equipped systems with the ability to interpret complex, nuanced queries, providing responses that go beyond simple keyword matches [7]. These models analyze the underlying intent of queries, making them significantly more effective than traditional retrieval systems. By leveraging these AI models, users can interact with large

datasets and receive more meaningful, context-aware responses, which opens up new possibilities for exploring and extracting insights from vast volumes of information.[8]

Generative AI goes beyond traditional document retrieval by synthesizing insights from multiple sources, allowing users to ask complex questions and receive real-time, human-like responses [9]. This dynamic approach connects scattered information, offering deep, contextually rich answers without manual navigation. In professional fields like research, law, and healthcare, generative AI enhances productivity by supporting multi-turn conversations and retaining context across queries, making it an invaluable tool for efficiently navigating and synthesizing information across various sectors [10].

Retrieval-Augmented Generation (RAG) frameworks combine generative AI with traditional retrieval methods, offering a hybrid solution for document interaction. RAG uses external knowledge sources to ensure that generated responses are both contextually relevant and grounded in factual information [11]. The retriever identifies pertinent documents, while the generator synthesizes answers based on the retrieved data and user query, overcoming the issue of factual inaccuracy in generative models. This framework is particularly useful in fields like academic research, legal analysis, and technical reviews, where precision and content generation are critical. RAG maintains conversational context across sessions, making it a powerful tool for dynamic, context-aware information extraction [12].

Conversational engagement transforms interactions by making them natural and intuitive. Unlike traditional systems where users must adapt to rigid interfaces or predefined commands like CTRL+F for searching, grep for pattern matching in text files, or find for locating files based on attributes, conversational systems allow users to communicate in a manner resembling human dialogue [13]. These classical methods, while functional, often require technical expertise and exact inputs, making them challenging for non-technical users. For instance, searching for specific clauses in a legal document using CTRL+F demands precise keyword knowledge, and any deviation in phrasing could render the search ineffective [14]. Conversely, conversational systems simplify such complex processes by allowing users to pose targeted questions about specific clauses and refine queries seamlessly, without losing context. The system, in turn, retains this conversational flow to provide tailored, precise responses [15]. This capability significantly reduces the effort required for tasks like searching, analyzing, or clarifying content. Moreover, conversational systems empower users to address ambiguities or refine their queries in real-time, ensuring a more personalized and effective interaction. By fostering a dynamic, back-and-forth dialogue, these systems improve accessibility and usability, bridging the gap between complex tasks and user-friendly solutions [16].

In addition to maintaining continuity, session management enhances interactions by supporting personalization and ensuring security. By analyzing previous interactions, chatbots can tailor their responses to individual user preferences and needs. This level of customization is invaluable in scenarios like customer service, where

remembering past issues or preferences leads to a more satisfying user experience. For example, a chatbot assisting in e-commerce can recommend products based on a user's previous purchases or searches, streamlining the shopping experience. However, personalization must be accompanied by robust security measures to safeguard user data. Advanced techniques such as encryption for session tokens, secure data storage, and session timeouts ensure that sensitive information remains protected throughout the interaction. This is particularly crucial in fields like finance or healthcare, where privacy breaches can have severe consequences. By combining personalization with secure session management, chatbots can deliver highly engaging, efficient, and trustworthy interactions, ensuring a balance between user convenience and data protection. This integration highlights the pivotal role of session management in enhancing the capabilities and effectiveness of conversational AI systems.

III. SYSTEM ARCHITECTURE

The system architecture integrates several key components to facilitate effective document interaction. PDF Pre-Processing involves extracting and segmenting text from uploaded PDF files using tools like PyPDFLoader, ensuring the documents are ready for further analysis. Embedding Generation using LLM models converts the extracted text into dense vector embeddings using models such as HuggingFace's all-MiniLM-L6-v2, enabling the system to capture the semantic meaning of document content. These embeddings are stored in a Vector Database for Semantic Search, where similarity-based retrieval is conducted. When a user inputs a query, the system retrieves the most relevant document chunks, significantly enhancing search accuracy over traditional keyword methods. Finally, Real-Time Response Generation involves using a generative model (e.g., Groq's "Gemma2-9b-It") to produce contextually appropriate, concise answers based on retrieved information. The architecture supports dynamic, real-time interaction with documents, making it effective for applications like academic research and legal review.

In the architecture of a Retrieval-Augmented Generation (RAG) system for document interaction, PDF pre-processing is a critical initial step. This stage involves converting uploaded PDF documents into a machine-readable format that can be parsed for content extraction. The system uses libraries such as PyPDFLoader to handle PDF files, as seen in the code provided. It extracts text from each PDF, handling potential issues such as multi-column layouts or embedded images that could affect accurate text extraction. Once extracted, the content is segmented into manageable chunks, which are then passed to the next stages for embedding generation. Proper PDF pre-processing ensures that the document data is structured, relevant, and ready for subsequent analysis, particularly in scenarios where large volumes of text must be processed in real time. This step lays the groundwork for effective document retrieval and interaction.

Transforming documents into embeddings using pre-trained language models like BERT or GPT involves a systematic process that converts textual data into dense numerical representations. These embeddings encode semantic and contextual information from the text, making it interpretable for machine learning tasks such as semantic search,

classification, and response generation. The process begins with text preprocessing, which prepares the raw document by tokenizing it into smaller units like words or subwords. For example, BERT uses WordPiece tokenization, ensuring compatibility with its vocabulary. Additional steps like removing special characters and standardizing the text help improve model compatibility. Since pre-trained models often have token length constraints (e.g., BERT can process up to 512 tokens per input), large documents are divided into smaller chunks, with each chunk processed independently to generate embeddings.

Once tokenized, the chunks are fed into the pre-trained language model, where the transformer architecture processes the input and generates embeddings. These embeddings are high-dimensional vectors that capture both the syntactic and semantic nuances of the text. Models like BERT produce context-sensitive embeddings, where each token's vector represents its meaning in context, while models like HuggingFace's all-MiniLM-L6-v2 offer compact embeddings optimized for efficiency and scalability. These chunk-level embeddings are then aggregated and stored in a vector database, such as ChromaDB or Pinecone, enabling efficient similarity searches using metrics like cosine similarity.

When a user submits a query, its embedding is generated using the same pre-trained model and compared with the stored embeddings to retrieve semantically relevant document chunks. These retrieved embeddings provide generative models like GPT with the necessary context to craft accurate and meaningful responses. For instance, a query about "payment terms" in a legal document would retrieve embeddings corresponding to relevant clauses, ensuring precision in the generated answer. This approach allows the system to bridge the gap between raw textual data and machine-readable formats, enabling advanced applications like conversational AI, intelligent content retrieval, and semantic search with remarkable efficiency and contextual understanding.

Preserving the semantic structure of textual data is a cornerstone of modern information retrieval systems, and embedding-based techniques excel in capturing this aspect. Unlike traditional keyword searches, which rely on exact word matches and often miss nuanced meanings, embeddings encode the semantic essence of a document or query. By transforming text into dense numerical vectors that represent its contextual and semantic meaning, embeddings enable a deeper understanding of the relationships within the content. This is crucial for applications requiring sophisticated information retrieval, as it ensures that relevant document sections are surfaced even when the wording of the query differs from the text.

Once these embeddings are generated, they are stored in a vector database like Chroma, which is tailored for handling high-dimensional data. The database supports fast and efficient similarity searches by indexing the embeddings and leveraging distance metrics such as cosine similarity to identify the closest matches. This approach allows the system to preserve the semantic relationships between document chunks and user queries. For example, even if a user asks about "payment terms" in a legal document without using the

exact phrasing found in the text, the system can retrieve the most relevant clauses because the embeddings capture the underlying meaning rather than just surface-level keywords.

This semantic preservation transforms the way information is retrieved, enabling systems to respond with contextually rich and accurate results. The use of vector databases enhances this process by maintaining the integrity of semantic structures during storage and retrieval. When a query is processed, its embedding is compared against stored embeddings to identify sections of the document that align with the query's intent. This capability is especially valuable in fields like legal analysis, academic research, and healthcare, where precision and contextual relevance are paramount. By focusing on semantics, embedding-based systems redefine information retrieval, making it more intuitive, responsive, and aligned with user needs.

Real-time response generation forms the interactive layer of this system architecture. After the relevant document embeddings are retrieved, the generative model, such as the Groq-powered "Gemma2-9b-It" mentioned in the code, synthesizes a human-like, context-aware response. The system uses a predefined prompt structure to guide the response generation, ensuring that it incorporates retrieved information and answers the user's query concisely. The real-time nature of this process is critical for user interaction, providing immediate feedback based on complex document queries. This architecture supports multi-turn conversations, preserving context through session history and enabling more in-depth interactions with documents. Real-time response generation, coupled with semantic search, enhances the system's ability to act as an intelligent assistant in diverse domains such as academic research, legal documentation, and healthcare reporting.

IV. METHODOLOGY

Document conversion to vector embeddings forms the foundation of our methodology, as it allows the system to process and understand textual content in a meaningful way, as shown in Fig.1. In this phase, documents, often in the form of PDFs, are transformed into vector representations using pre-trained language models. Specifically, the Hugging Face Embeddings module from Lang Chain is employed to generate high-dimensional embeddings. These embeddings capture the semantic structure of the text, making it easier for the system to analyze and retrieve relevant information. For instance, by utilizing models like BERT or GPT, each section of the document is embedded into vectors that encode not only the words but also the relationships between them.

To ensure efficient processing, especially with larger documents, the Recursive Character Text Splitter is used to divide the text into smaller segments. This step is crucial for optimizing the embeddings, as processing smaller text chunks helps preserve contextual relevance and prevents the model from being overwhelmed by too much data at once. The generated embeddings are then stored in the Chroma vector database, which supports efficient semantic search. This allows the system to later retrieve documents or portions of text based on their meaning, rather than relying on simple keyword matching. This process is critical in ensuring that

users receive accurate and contextually relevant responses when interacting with the system.

The `create_history_aware_retriever` and `create_retrieval_chain` modules play essential roles in mapping user queries to the pre-generated embeddings, ensuring that responses are both contextually accurate and relevant to the ongoing conversation. The `create_history_aware_retriever` module is specifically designed to account for the history of the user's interactions, enabling the system to incorporate context from previous queries into the retrieval process. This feature is crucial in multi-turn dialogues, where the user's questions may build upon earlier inquiries.

By maintaining a history of the conversation, the system can tailor responses more effectively, providing answers that reflect the user's evolving needs and the broader context of the discussion. For instance, if a user inquires about payment terms in a legal document and later asks for related clauses, the system, using this module, can retrieve answers that are informed by both the current query and the conversation history, ensuring continuity and relevance in the interaction.

On the other hand, the `create_retrieval_chain` module is responsible for processing the user query and matching its semantic meaning with the relevant document sections stored as embeddings in the Chroma vector database. This module serves as the crucial link between the user's natural language input and the pre-processed, vectorized data. When a query is submitted, the retrieval chain transforms it into an embedding, which is then compared against the embeddings stored in the vector database.

The system uses similarity metrics like cosine similarity to identify the closest matching document segments, ensuring that the response is aligned with the meaning behind the query. The retrieval chain ensures that even complex, multi-step queries are answered with precise and contextually appropriate content, making the entire process seamless for the user.

Together, these two modules create a robust retrieval system. While the history-aware retriever ensures that the context is continuously maintained and that responses are coherent across multiple interactions, the retrieval chain processes the query and retrieves the most relevant document sections based on their semantic content. The combination of these modules enables an efficient, context-aware system that can handle sophisticated user queries, providing accurate and meaningful responses with a deep understanding of the conversation's flow.

This integration of the `create_history_aware_retriever` and `create_retrieval_chain` modules significantly enhances the user experience by ensuring that responses are not only contextually accurate but also dynamically adapted to the flow of the conversation. By leveraging both the historical context and semantic search capabilities, the system can efficiently handle complex, multi-step queries that require understanding across various stages of the dialogue. This advanced approach minimizes the need for users to repeat themselves and allows the system to provide more relevant, precise answers. Moreover, as the conversation evolves, the system becomes increasingly adept at tailoring its responses to meet the user's specific needs, further streamlining the process and improving overall interaction quality. This functionality is especially valuable in industries like legal analysis, customer service, and healthcare, where understanding the broader context of inquiries is crucial for delivering effective and accurate responses.

Session management is vital for ensuring coherent, contextually aware conversations in chatbots. Using modules like Chat Message History and Base Chat Message History from Lang Chain, chatbots track and store interactions, allowing them to recall previous queries and responses. This enables smooth dialogue, where the system can reference past exchanges for accurate, informed replies.

The Runnable With Message History module further enhances by preserving message history during response generation, ensuring that interactions remain connected. Overall, effective session management significantly improves dialogue flow, user satisfaction, and engagement by making conversations contextually rich and intuitive.

Data security is an essential aspect of any system that processes user queries, stores sensitive information, and interacts with confidential documents. This is especially true for applications that handle legal contracts, personal health records, or financial data. One of the fundamental security best practices is ensuring that data is encrypted both in transit and at rest. Encryption in transit protects the data as it moves across the network, ensuring that any information exchanged between the client and server is unreadable to unauthorized parties. This can be achieved through secure protocols like

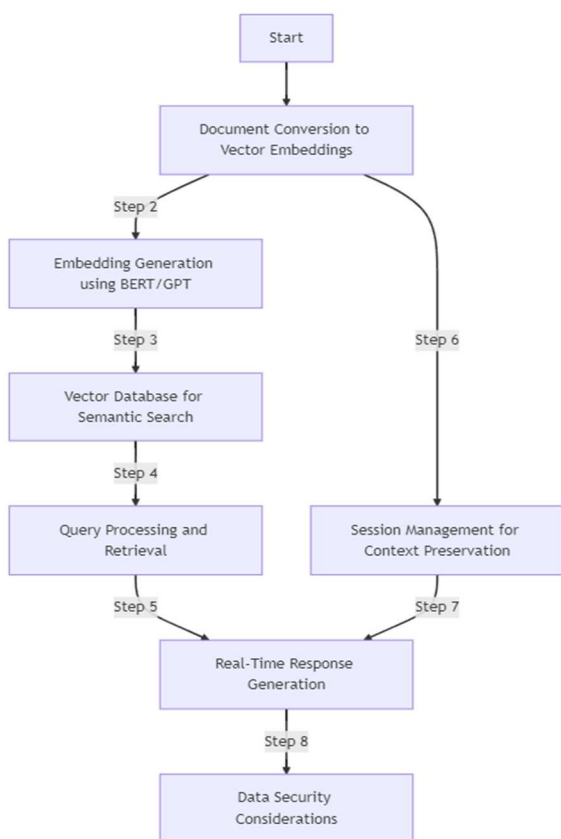


Fig 1: Proposed Model

HTTPS (TLS/SSL), which ensures that the data remains encrypted while being transferred. On the other hand, encryption at rest protects stored data, making it unreadable even if the physical storage media is compromised. This can be accomplished using industry-standard encryption algorithms like AES-256, which ensures that even if an attacker gains access to the storage system, they cannot decipher the sensitive data without the decryption key. Both of these encryption techniques are critical in safeguarding user data and preventing unauthorized access, ensuring that the integrity and confidentiality of the system are maintained.

Environment variables play a crucial role in enhancing the security of an application, particularly when managing sensitive information such as API keys and configuration settings. Instead of hardcoding sensitive data directly into the application's source code, developers store this information in environment variables. This practice helps reduce the risk of exposure, especially in scenarios where code may be shared publicly or pushed to version control systems. The `os` module in Python, for instance, allows the application to securely access these environment variables without exposing them in the codebase. By utilizing environment variables, sensitive data like API keys, database credentials, and encryption keys are protected from unauthorized access, as they are stored in secure locations like `.env` files or dedicated key management systems. This approach ensures that even if the source code is exposed, sensitive information remains protected.

Session expiration policies and monitoring for unusual activity are also critical components of data security in systems that store and process user interactions. Session expiration ensures that, after a period of user inactivity, the system automatically terminates the session, reducing the risk of unauthorized access. For example, if a user leaves their session unattended for an extended period, the system could automatically log them out or require re-authentication before granting access to sensitive data. This is especially important for applications dealing with highly sensitive information, as it prevents attackers from exploiting open sessions. Additionally, session expiration can be combined with other security mechanisms, such as two-factor authentication (2FA), to provide an extra layer of protection. For systems that require continuous interactions, such as in legal or healthcare contexts, secure session management can allow users to resume their activities without compromising security. By using token-based authentication or other session management tools, the system can securely store session data and grant access only to authorized users. Furthermore, monitoring for unusual activity, such as multiple failed login attempts or queries that deviate from typical patterns, is essential for detecting potential security breaches. Real-time monitoring and logging of all system activities help identify suspicious behaviors, allowing security teams to act quickly and prevent further exploitation. By combining robust session expiration policies with constant monitoring, the system can maintain the highest levels of security and ensure that user data remains safe throughout the entire interaction process.

Academic researchers often need to reference specific sections or extract insights without manually navigating the entire document. The system, leveraging Generative AI with

the Retrieval-Augmented Generation (RAG) framework, significantly streamlined this process by enabling users to input natural language queries and receive immediate, contextually relevant responses.

V. CHALLENGES AND RESULTS

One of the primary challenges in implementing Generative AI systems, especially those integrated with frameworks like Retrieval-Augmented Generation (RAG), is maintaining accuracy in responses. The system relies on converting documents into embeddings, which are essentially vectorized representations of the document's semantic meaning. However, the accuracy of the system can be affected by several factors, including the quality of the embeddings, the complexity of user queries, and the inherent limitations of the transformer-based models (e.g., BERT, GPT) used in generating these embeddings.

Maintaining response accuracy is critical, especially in sensitive domains like legal document review or healthcare reporting, where incorrect interpretations could lead to severe consequences. The system might misinterpret complex legal clauses or medical terminologies if the embeddings do not sufficiently capture the intricacies of the language. For example, in the healthcare domain, medical reports may include nuanced jargon that might be too complex for the transformer models to encode accurately. This can lead to instances where the system provides incorrect or imprecise responses, impacting the overall reliability of the tool.

To mitigate this, several techniques can be employed. One potential solution is to enhance the pre-processing phase by incorporating domain-specific language models. For instance, in the case of legal documents, models trained specifically on legal text can be utilized to improve accuracy. Similarly, healthcare applications can benefit from fine-tuning the model on medical datasets to enhance the system's understanding of medical terminology. Another alternative is to introduce multi-step query refinement, where the system asks clarifying questions before delivering a final response to ensure that it is interpreting the user's query correctly.

Furthermore, session management, which plays a key role in maintaining conversational context, poses its own challenges in accuracy. While session management helps in preserving the flow of conversations, ensuring that follow-up questions are answered with context, it can sometimes propagate misunderstandings if the initial interpretation was incorrect. This cascade of errors can lead to an accumulation of inaccuracies as the session progresses. Incorporating feedback mechanisms that allow users to correct or refine the system's responses could help improve overall accuracy. This interactive correction would enable the AI to better understand user intent and context over time.

Finally, a challenge that cannot be overlooked is the model's tendency to generate hallucinations—providing responses that are syntactically plausible but factually incorrect. This is a known limitation of current transformer-based models, which often predict responses based on patterns in data rather than genuine understanding. A robust post-processing phase, involving human-in-the-loop verification, could be explored as an alternative for use cases where response accuracy is critical. The Table I below shows a comparative study of Retrieval-Augmented Generation (RAG) with the existing Generative models like Bert and GPT.

TABLE I. COMPARITIVE CHART OF EXISTING AND PROPOSED MODELS

Parameters	Retrieval-Augmented Generation (RAG)	BERT Model	GPT Model
Task / Use Case	Open-domain Question Answering Method (QA)	Sentence-level Understanding	Text Generation
SQuAD (F1 Score)	76.0% (RAG-Token)	89.3% (BERT-large)	~73.0% (GPT-3)
Natural Questions (F1 Score)	66.0% (RAG-Token)	-	~65.0% (GPT-3)
Natural Questions (EM Score)	50.0% (RAG-Token)	-	~60.0% (GPT-3)
Text Completion (Perplexity)	-	-	20.0 (GPT-3)
Inference Time	Slower (~4-5 seconds for long answers)	Fast (~1-2 seconds)	Fast (~2-3 seconds, varies with prompt length)

VI. CONCLUSION

This research highlights the transformative potential of combining Generative AI with the Retrieval-Augmented Generation (RAG) framework to improve document-based interactions across various fields, such as academic research, legal reviews, and healthcare reporting. By leveraging transformer models like BERT and GPT for embedding generation, the system converts large, complex documents into semantically enriched vectors, allowing for efficient document navigation through semantic search. This approach enables users to retrieve relevant information quickly and accurately, overcoming the limitations of traditional keyword-based search methods. Real-time response generation further enhances the user experience by tailoring responses based on the historical context of interactions, ensuring more contextually relevant answers. This efficiency is particularly crucial in fields like law and healthcare, where precision is essential for decision-making and time management.

Despite its effectiveness, the system faces challenges that require attention in future work. One key limitation is the transformer models' understanding of domain-specific terminology. While models like BERT and GPT perform well in general language tasks, they can misinterpret specialized terms or fail to grasp the nuances of fields like law and medicine, potentially leading to inaccuracies in information

retrieval and response generation. Another issue is the occurrence of model hallucinations, where the AI generates contextually inaccurate or misleading content. These challenges emphasize the need for improvements in domain-specific fine-tuning and better handling of complex, multi-step queries, which would improve both the accuracy and reliability of the system in specialized fields.

REFERENCES

- [1] A. Elgammal, B. Liu, and J. Song, "Creative adversarial networks: Generating art by learning about styles and deviating," *ACM Trans. Graphics*, vol. 36, no. 4, pp. 1–13, 2017.
- [2] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, pp. 140:1–140:67, 2020. [Online]. Available: <http://jmlr.org/papers/v21/20-074.html>
- [3] Foster, C., & Bell, S. (2024). AI-Driven Multimodal Platforms for Content Personalization. *Journal of Information Technology*, 39(1), 15-31.
- [4] KAON Ontology and Semantic Web Infrastructure <http://kaon.semanticweb.org>
- [5] J. Smith and P. Taylor, "Personalization in AI-generated content for marketing campaigns," *J. Interactive Marketing*, vol. 54, pp. 122–137, 2024
- [6] N. Voskarides, D. Li, P. Ren, E. Kanoulas, and M. de Rijke, "Query resolution for conversational search with limited supervision," *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020.
- [7] N. Anderson and K. Roberts, "Improving creative processes using AI-driven content generators," *Creativity Innov. Manage.*, vol. 33, no. 1, pp. 45–60, 2024.
- [8] R. Baeza-Yates, B. Ribeiro-Neto, "Modern Information retrieval", Addison-Wesley, 1999.
- [9] S. Sukhbaatar, A. Szlam, J. Weston, and R. Fergus, "End-to-end memory networks," in *Advances in Neural Information Processing Systems 28: Annual Conference on Neural Information Processing Systems 2015, December 7-12, 2015, Montreal, Quebec, Canada*, 2015, pp. 2440–2448. [Online]. Available: <http://papers.nips.cc/paper/5846-end-to-end-memory-networks>
- [10] S. Lin, J. Yang, R. Nogueira, M. Tsai, C. Wang, and J. Lin, "Conversational question reformulation via sequence-to-sequence architectures and pretrained language models," *CoRR*, vol. abs/2004.01909, 2020.
- [11] S. Yu, J. Liu, J. Yang, C. Xiong, P. N. Bennett, J. Gao, and Z. Liu, "Few-shot generative conversational query rewriting," in *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25-30, 2020. ACM, 2020*, pp. 1933–1936. [Online]. Available: <https://doi.org/10.1145/3397271.3401323Y>.
- [12] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual Web search engine. In Ashman and Thistlewaite [2], pages 107–117. Brisbane, Australia
- [13] Wu, Zhuohao, Danwen Ji, Kaiwen Yu, Xianxu Zeng, Dingming Wu, and Mohammad Shidujaman. "AI creativity and the human-AI co-creation model." In *Human-Computer Interaction. Theory, Methods and Tools: Thematic Area, HCI 2021, Held as Part of the 23rd HCI International Conference, HCII 2021, Virtual Event, July 24–29, 2021, Proceedings, Part I 23*, pp. 171-190. Springer International Publishing, 2021.
- [14] Rezwana, Jeba, and Mary Lou Maher. "Designing creative AI partners with COFI: A framework for modeling interaction in human-AI co-creative systems." *ACM Transactions on Computer-Human Interaction* 30, no. 5 (2023): 1-28.
- [15] Das, Payel, and Lav R. Varshney. "Explaining artificial intelligence generation and creativity: Human interpretability for novel ideas and artifacts." *IEEE Signal Processing Magazine* 39, no. 4 (2022): 85-95.